

A Multimodal Framework for Early-Stage Crop Disease Diagnosis Using Visual and Environmental Data

Abstract—Early detection of crop diseases is critical for reducing yield losses and enabling timely agricultural intervention. However, diagnosis based on visual symptoms alone is unreliable when early stage symptoms are subtle, field acquired images are degraded, or environmental conditions shift rapidly. Because image only deep learning models capture only visual information, they cannot account for environmental drivers of disease progression such as temperature and humidity. This thesis addresses that limitation by combining crop imagery with IoT-based environmental sensing through multimodal data fusion. The framework uses classical neural networks to produce compact representations of the visual and sensor data: a fine tuned MobileNetV3-Large backbone provides a frozen 1280 dimensional image embedding, and a compact MLP over 54 statistical sensor features provides a frozen 64 dimensional environmental embedding. These are concatenated and passed through classical fusion heads, evaluated across bottlenecked and unconstrained widths on a large, highly imbalanced agricultural dataset (25 crop-disease-stage classes). The models are assessed on clean data and under controlled test-time image corruption, sensor corruption, and combined image-and-sensor corruption. The results show that multimodal fusion consistently outperforms single-modality baselines on clean data, with our unconstrained fusion model achieving a higher Macro F1, 98.80% accuracy, and 98.12% early-stage recall. Crucially, fusion also exhibits superior robustness, degrading far more gracefully than unimodal models when either or both input channels are corrupted. These findings establish a strong classical multimodal benchmark and robustness baseline for crop disease stage classification under realistic field deployment conditions.

I. INTRODUCTION

The global population is projected to reach 9.7 billion by 2050, demanding a sharp rise in food production even as pathogens and pests cause heavy annual crop losses. Because traditional visual inspection typically detects disease only after it is well-established, precision agriculture has increasingly turned to AI to shift from reactive treatment to proactive, early-stage diagnosis. Image-based deep learning (CNNs, Vision Transformers) is effective but vulnerable to field degradation—poor lighting, blur, and acquisition noise—and blind to the environmental drivers of disease such as temperature, humidity, and dew point, which visual data alone cannot capture. To address this, we propose a multimodal framework that fuses high-resolution crop imagery with IoT sensor telemetry, improving early-stage detection and withstanding real-world visual and sensor noise. To this end, we evaluate lightweight unimodal baselines (MobileNetV3 for images, MLP for sensors) that establish strict performance references, then build on them with a classical multimodal fusion architecture that

classifies 25 crop-disease-stage categories and outperforms both single-modality classifiers. We further conduct a corruption robustness study against realistic image and sensor degradation, showing that robustness is intrinsic to the fusion architecture. Finally, we establish a parameter-matched classical performance ceiling and bottleneck interface, providing the baseline for our future Variational Quantum Circuit (VQC) fusion head under NISQ constraints.

II. LITERATURE REVIEW

Ref No	Dataset	ML Models	Accuracy
[1]	PlantVillage tomato	ToLeD (custom CNN)	91.2%
[2]	PlantVillage	DENN (ensemble)	99.99%
[3]	PlantVillage + PlantDoc	ViT + MoE	+20% over ViT
[3]	Watermelon (UAV/benchtop)	NN + spectral indices	>90% severe; 62% early
[4]	Jamun leaf + IoT	MLP-CNN fusion	~95%
[5]	Wheat + env. data	Multimodal fusion	96.5%, F1 95.9%, 180 ms
[2]	Drone + IoT	AgriFusionNet	High eff./acc
[6]	205k images + text	PlantIF (graph fusion)	96.95%

Table 1: Performance summary of representative models

Classical ML and deep learning are highly effective at processing crop imagery, but a single visual modality is fundamentally limited: the earliest, pre-symptomatic disease stages often leave no visible lesion, and field images degrade under poor lighting, blur, occlusion, and noise. Environmental variables — temperature, humidity, soil moisture, and leaf wetness — drive disease onset and are captured continuously by IoT sensors, yet they are invisible in an image. This motivates multimodal fusion: combining visual evidence with environmental context to classify disease stages more reliably than either modality alone. Fusion is non-trivial, though — image and sensor streams differ in scale, structure, and temporal resolution (Li and Tang, 2025) [7], naive strategies assume linear relationships, and a high-dimensional image embedding can suppress the low-dimensional sensor signal. Remote sensing had already shifted diagnosis from subjective inspection toward quantitative multispectral, thermal, and hyperspectral collection that reveals physiological stress before visible symptoms (Zhang et al., 2019; Wu et al., 2026) [6], [8],

establishing that a diagnostically useful signal exists outside the plain RGB image.

A. Classical Machine Learning and Its Limits

High-resolution sensing outpaced manual analysis, triggering the classical ML era. Early work used vegetation indices and hard thresholds that flagged general stress but not specific pathogens (Abdulridha et al., 2022) [9]. Handcrafted-feature classifiers — SVM, KNN, Random Forest — depended entirely on manual feature engineering (Abbas et al., 2023) [10] and suffered from inter-class spectral overlap (drought vs. fungal infection look similar). KNN degraded in high-dimensional data (64—83% depending on metric; Uddin et al., 2022 [11]), and SVMs handled severe stages well (>90%) but collapsed on asymptomatic early infections (~62%; Abdulridha et al., 2022 [9]). These early-stage failures — subtle, pixel-scale spectral shifts rather than distinct shapes — motivated the move to representation learning on raw data.

B. Deep Learning: CNNs, Ensembles, and Transformers

CNNs automated feature extraction and dominate image-based detection (Araújo et al., 2023) [12], but off-the-shelf backbones need adaptation: Jiang et al. (2020) [13] raised ResNet-50 to 98.0% via Leaky-ReLU and larger kernels, while a simple custom CNN beat heavy pretrained models on grape disease (Math and Dharwadkar, 2022) [14], showing depth yields diminishing returns on low-variance crop data. For field deployment, efficiency dominates: lightweight ToLeD (Agarwal et al., 2020) [1] and YOLO variants (Redmon et al., 2016 [15]; Wang et al., 2021 [16]; Amarasingam et al., 2022 [17]) trade some accuracy for edge-deployable speed and size, outperforming data-hungry transformers like DETR on small datasets. To push robustness further, ensembles combine diverse models — DENN reached 99.99% but at heavy compute cost (Vallabhajosyula et al., 2022) [18], while lighter variants target IoT constraints (ECPRC; Nagasubramanian et al., 2021 [19]). Vision Transformers add global context: MoE-ViT recovers field-condition accuracy (+20%; Salman et al., 2025 [3]). Across all of this, two constants remain — rising computational cost, and the fact that every model reasons over a single modality.

C. IoT-Driven Multimodal Fusion

Static imaging cannot capture the unseen environmental drivers of pathogen growth, so IoT and fusion shift the goal from diagnosing a symptom to understanding the system. Timeliness of environmental data is the key intervention factor (Atzberger, 2013 [20]), and IoT—big-data pipelines enable proactive management despite latency and infrastructure costs (Alahmad et al., 2023 [21]). The central finding across recent work is complementarity: image methods miss early disease for lack of physical signs, while sensor methods cannot localise it spatially (Garg et al., 2025) [4]. Environmental readings thus act as probabilistic priors that disambiguate visual evidence — a marginal lesion becomes a likely fungal

infection under high leaf-wetness. Concatenation-based MLP-CNN fusion reached ~95% on Jamun (vs. 87–90% single-modality; Garg et al., 2025 [4]) and 96.5% on wheat at 180 ms (Xu and Wang, 2025 [5]); lightweight multi-source models (AgriFusionNet; Albahli, 2025 [2]) and graph-based image—text fusion (PlantIF, 96.95%; Wu et al., 2026 [6]) extend the idea at rising complexity. Notably, almost all of this work reports clean-data accuracy only — behaviour under a degraded modality is largely unexamined.

III. DATASET

The Dacon LG Crop Disease Diagnosis dataset was selected for its scale, diversity, and explicit inclusion of disease progression stages. It combines high-resolution crop images with greenhouse IoT environmental sensor data.

The dataset covers 6 crop types and 12 disease codes across 4 progression stages (Healthy, Early, Mid, and Severe), targeting early-stage recognition. This yields 25 valid crop-disease-stage label combinations across 57,673 total samples.

A. Class Imbalance

The dataset exhibits significant natural class imbalance reflective of real-world agricultural conditions (Table II). Samples are heavily skewed toward Healthy and Early stages, while the Severe stage accounts for only 400 samples (~0.7%).

Progression Stage	Training Images	Validation Images	Test Images	Total Images
Healthy (Stage 0)	32,081	3,564	3,944	39,589
Early (Stage 1)	10,844	1,205	1,327	13,376
Mid (Stage 2)	3,481	387	440	4,308
Severe (Stage 3)	307	37	56	400
Total	46,713	5,193	5,767	57,673

Table 2: Dataset distribution across progression stages.

To handle this imbalance, we adopt two strategies: using Weighted Cross-Entropy loss during training to prevent majority-class bias, and emphasizing Macro F1 and early-stage recall alongside accuracy as evaluation metrics.

B. Data Pre-processing

Images were resized to 224×224 , and all modalities were standardized prior to modeling. Rather than modeling the raw sensor time-series directly, each record was summarized into 54 fixed statistical features (mean, standard deviation, min, max, median, and last value across 9 environmental variables) for the MLP architecture.

To prevent data leakage, a single fixed train/validation/test split was maintained across all experiments.

C. Augmented Test Set

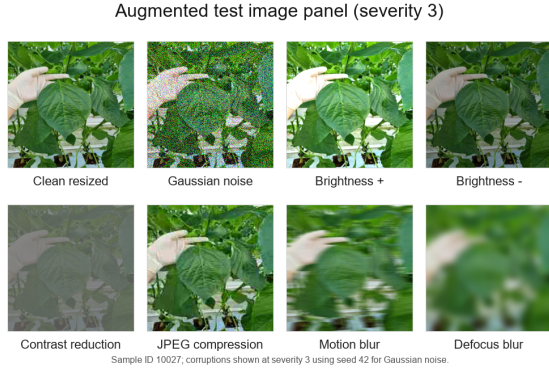


Fig. 1. Clean test image and the seven image-corruption variants used in the robustness evaluation.

The augmented test set is used only for robustness evaluation, and every corruption alters the input without changing the underlying class label. The protocol adapts the ImageNet-C common-corruption benchmark and extends it beyond the image channel to the sensor modality. Three families are applied: seven image corruptions (Gaussian noise, brightness increase/decrease, contrast reduction, JPEG compression, motion blur, and defocus blur), three sensor corruptions on the 54-dimensional feature vector (sensor noise, feature dropout, and sensor drift), and joint corruptions that degrade both channels at once using a standard pair (Gaussian noise + sensor noise) and a worst-case pair (defocus blur + drift). Each corruption is applied at severity levels 2, 3, and 5 to span mild to aggressive degradation, with three random seeds (42, 123, 999) for stochastic corruptions and a single materialised set for deterministic ones.

IV. METHODOLOGY

The methodology follows a systematic experimental design to ensure equitable comparisons across architectures.

A. Model Specification

1) *Image Encoder Baseline (E0)*: The E0 model serves as the foundational computer vision baseline. It uses a MobileNetV3-Large backbone, pre-trained and fine-tuned on the agricultural dataset. After fine-tuning, the classification head is stripped away and the model is frozen, functioning as a fixed feature extractor that outputs a dense 1280-dimensional embedding from its penultimate layer for every input image. This frozen embedding feeds directly into the multimodal fusion stages.

2) *IoT Environmental Baseline (E1)*: The E1 model processes the structured environmental telemetry data. Its input consists of the 54-dimensional statistical feature vectors. The architecture is a compact Multi-Layer Perceptron. For multimodal fusion, the 64-dimensional hidden representation produced at the penultimate layer is used as the frozen environmental embedding. No residual connection is used in the E1 model. The E1 architecture has 17,305 trainable parameters.

3) *Fusion Model (E2)*: The E2 model establishes the performance ceiling for multimodal fusion. It projects the 1344-dimensional combined vector through a progressive bottleneck. The full architecture is:

$\text{Linear}(1344, 128) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(128, k) \rightarrow \tanh \rightarrow \text{Linear}(k, k, \text{bias}=\text{False}) \rightarrow \tanh \rightarrow \text{Linear}(k, 25)$

Here, k represents the bottleneck dimension (4 or 8). The inner $\text{Linear}(k, k, \text{bias}=\text{False})$ transformation learns non-linear interactions across the highly compressed feature space before expanding back into 25-class logits. The E2 model with an 8-dimensional bottleneck (E2-matched-8) has 173,737 trainable parameters.

A second, unconstrained classical fusion head — referred to throughout as E2-128-25 — is also evaluated. It removes the bottleneck entirely and instead classifies the fused 1344-dimensional vector at full width: it removes the inner bottleneck, with 175,641 trainable parameters. Because it is not restricted to the narrow bottleneck budget, E2-128-25 represents the unconstrained performance ceiling; it is used as the upper reference point in the corruption-robustness study, against which the bottlenecked head (E2-8) is compared.

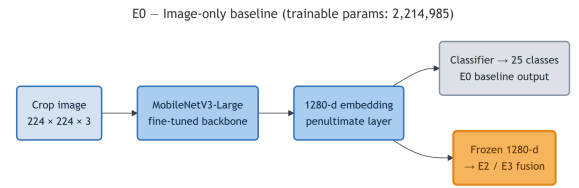


Fig. 2. Architectural block diagram of the Fusion Model (E2).

Experiment	Architecture Description	Modality
E0 Baseline	MobileNetV3-Large + classifier(25)	Image only
E1 Baseline	IoT MLP [128, 64] + classifier(25)	IoT only
E2 Fusion	Frozen E0 + Frozen E1 + Classical MLP head + k + classifier(25)	Multimodal

Table 3: Summary of proposed architectures

B. Experimental Strategy

The comparison is cumulative: each stage reuses the previous stage’s frozen outputs, so any performance difference is attributable to the architectural component added at that stage rather than to changes in the encoders or the data split. A single fixed train/validation/test split is reused throughout to prevent data leakage. First, the two unimodal baselines were trained and tested separately to establish per-modality reference points—E0 for the image channel, E1 for the sensor channel—then frozen and used purely as feature extractors, supplying a 1280-dimensional image embedding and a 64-dimensional sensor embedding respectively. Next, these frozen embeddings were concatenated and passed through the classical fusion head to train E2 at both constrained bottleneck

widths (E2-4, E2-8) and the unconstrained full width (E2-128-25). Comparing E2 against E0/E1 on clean data isolates the benefit of fusion; comparing the bottlenecked heads against the unconstrained one isolates the cost of compressing the fused representation. Since all variants share identical frozen encoders and differ only in the fusion head, the comparison is controlled and parameter-matched. Finally, all four models (E0, E1, E2-8, E2-128-25) were re-evaluated on corrupted versions of the held-out test set without retraining. Each unimodal model is the reference for its own corruption type (E0 image, E1 sensor), while the fusion heads are tested on image, sensor, and joint corruption. The primary robustness metric is Macro-F1 drop from each model’s own clean baseline, and a paired difference-of-drops bootstrap (95

V. RESULT ANALYSIS

A. Performance Evaluation

The primary goal was to measure the diagnostic gain of multimodal fusion over single-modality baselines; the secondary goal was to test whether that gain survives realistic test-time degradation.

1) *Single-Modality Baselines (E0 and E1)*: The image-only baseline (E0) reached 98.21% accuracy but captures no environmental disease precursors. The sensor-only baseline (E1) provided environmental context but lacked spatial symptom information, giving much weaker severe-stage recall (48.21%).

2) *Multimodal Fusion and Bottleneck Evaluation (E2)*: The 4D bottleneck (E2-matched-4) reached 92.42% accuracy but collapsed on severe-stage recognition (25.00% severe-stage recall), the compression breaking key stage-separation boundaries. Widening to 8D restored them: E2-matched-8 was the best predictive classifier at 98.75% accuracy and 0.9733 Macro F1, improving early-stage recall over both baselines and restoring severe-stage recall to 80.35% (matching, not exceeding, E0). The unconstrained head (E2-128-25) performed on par on clean data (98.80% accuracy, 0.9734 Macro F1, 98.12% early-stage recall, 78.57% severe-stage recall), confirming that relaxing the bottleneck carries no clean-data penalty. Calibration, however, splits along the bottleneck axis: E2-128-25 and both unimodal baselines are well calibrated (ECE 0.008—0.017), whereas the bottlenecked heads are badly overconfident (ECE 0.2483 at 8D, 0.3365 at 4D).

Model	Architecture	Test Accuracy	Test Macro F1	Early-Stage Recall	Severe-Stage Recall	Test ECE
E0	Image Only	98.04%	0.9639	98.04%	80.35%	0.0078
E1	IoT-only	94.36%	0.8834	93.97%	48.21%	0.0165
E2-matched-4	Fusion(4D Bottleneck)	92.42%	0.6614	97.13%	25.00%	0.3365
E2-matched-8	Fusion(8D Bottleneck)	98.12%	0.9733	98.41%	80.35%	0.2483
E2-128-25	Fusion(Unconstrained, 128)	98.80%	0.9734	98.12%	78.57%	0.0076

Table 4: Held-out test performance of E0, E1, the matched-bottleneck heads (E2-matched-4/8), and the unconstrained head (E2-128-25).

3) *Corruption Robustness Evaluation*: All models were evaluated on the corrupted test sets without retraining, at severities 2, 3, and 5; the primary metric is Macro-F1 drop from each model’s own clean baseline, and a severity-0 check returned zero delta for every model. One ordering holds across every image, sensor, and joint condition tested: E2-128-25 > E2-8 > unimodal.

Image corruption: Under image corruption, the image-only baseline fails exactly where field deployment is most likely to strain it: photometric shifts are benign (brightness costs E0 only 0.012 mean Macro-F1), but structural and stochastic degradation is catastrophic—Gaussian noise and defocus blur both drive E0 to near-random (0.010 and 0.012 at severity 5). Both fusion heads remain functional under the same inputs (0.324 and 0.433 under Gaussian noise), which is only possible if the sensor embedding still carries class-discriminative signal once the image is uninformative.

The sensor channel shows the mirrored pattern: E1 is non-functional from severity 2 under drift (0.132), while E2-128-25 retains 0.866. Drift is both the most damaging corruption and the one where representation width matters most—mean drop falls from 0.583 to 0.338 when the bottleneck is removed. Under joint corruption the ordering survives with both channels degraded (standard pair, severity 2: E2-8 0.538, E2-128-25 0.697), establishing the advantage as structural rather than dependent on one clean channel.

Sensor corruption. Same ordering (E1 < E2-8 < E2-128-25). The sensor channel mirrors the image case: the sensor-only baseline fails under exactly the degradation a field deployment invites, while fusion absorbs it. Sensor noise already costs E1 0.231 Macro-F1 at severity 2, yet the fusion heads barely register it (E2-8 drops only 0.015, $\sim 15\times$ less). Drift is the decisive case — by shifting the whole feature distribution rather than adding local noise, it leaves E1 non-functional from severity 2 (0.132), but E2-128-25 still retains 0.866, and it is here that representation width pays off most, nearly halving the mean drop (0.583 $\rightarrow$ 0.338, the study’s single largest robustness gain). Fusion can only stay diagnostic once the sensor stream is corrupted because the frozen image branch continues to supply class-discriminative signal.

Joint corruption (both modalities). Two pairs tested. When both channels degrade at once, the fusion advantage no longer depends on one clean modality propping up the other — it survives with no clean input at all. Under the standard pair (Gaussian + sensor noise) the ordering holds at every severity, E2-128-25 leading E2-8 throughout (0.697 vs 0.538 at severity 2). The worst-case pair (defocus blur + drift) pushes both heads toward the failure floor, near-random by severity 3, yet E2-128-25 still leads wherever any signal remains (0.328 vs 0.163 at severity 2). That the gap persists with both modalities corrupted confirms the robustness is structural — a property of the fused representation itself, not of either channel staying intact.

Corruption	E2-8 mean F1 drop	E2-128-25 mean F1 drop
<i>Image</i>		
Brightness increase	0.006	0.003
Brightness decrease	0.016	0.007
JPEG compression	0.167	0.113
Contrast reduction	0.187	0.108
Motion blur	0.216	0.143
Gaussian noise	0.490	0.396
Defocus blur	0.407	0.230
<i>IoT (sensor)</i>		
Feature dropout	0.175	0.087
Sensor noise	0.211	0.130
Sensor drift	0.583	0.338
<i>Joint (both modalities)</i>		
Gaussian noise + sensor noise	0.681	0.582
Defocus blur + drift	0.915	0.816

Table 5: Mean Macro-F1 drop (severities 2/3/5) under image, IoT, and joint corruption for the two fusion variants (smaller is better).

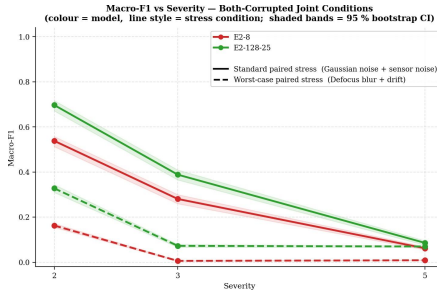


Fig. 3. Macro F1 vs. severity under joint corruption for the fusion heads (E2-8, E2-128-25), for the standard and worst-case pairs.

B. Analysis of Design Solutions

1) *Bottleneck Sweep and Severe-Stage Confusion*: At 4D, the fusion model captured early- and mid-stage features but lost severe-stage features, compressing them into the early/healthy classes. Expanding to 8D restored 80.35% severe-stage recall.

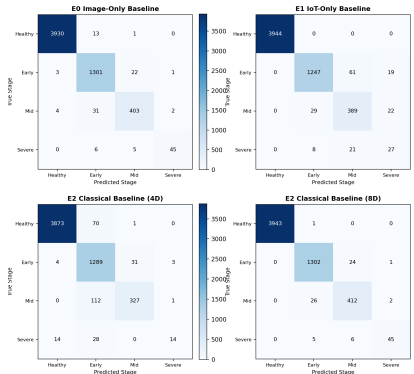


Fig. 4. Stage-level confusion matrices (held-out test set) for E0, E1, and fusion (E2 4D/8D).

2) *IoT Representation and Modality Balance*: Two design choices are validated by the results. First, representing each sensor stream by 54 fixed summary statistics (rather than a

temporal model) preserves the diagnostic signal: the 17,305-parameter E1 MLP reaches 0.8834 Macro F1, 94.36% accuracy, and 93.97% early-stage recall, though its 48.21% severe-stage recall marks the sensor modality as a complementary prior, not a standalone diagnostic. Second, concatenating a 64-d sensor embedding with a 1280-d image embedding did not let the visual signal swamp the sensor branch: on clean data, fusion raises Macro F1 (0.9639 $\rightarrow$ 0.9733), and under severity-5 Gaussian noise (E0 $\rightarrow$ 0.010) the fusion heads retain 0.324 (E2-8) and 0.433 (E2-128-25) — impossible unless the sensor embedding remained an active, weighted part of the representation.

3) *Robustness Advantage of Multimodal Fusion*: Two findings separate the value of fusing modalities from the value of preserving representation width: E2-8 degrades less than the relevant unimodal baseline, and E2-128-25 degrades less than E2-8. This is statistically robust — the paired difference-of-drops bootstrap (95% CI) excludes zero in 44 of 48 image/sensor conditions for E2-8 and 46 of 48 for E2-128-25, the few non-significant cases confined to low-severity brightness shifts and (for E2-8) sensor drift at severity 5, where both E1 and E2-8 have collapsed. The mechanism: the frozen IoT embeddings supply class-discriminative context orthogonal to pixel values (bounding image-corruption loss), while the frozen image branch supports the model under sensor corruption. The bottleneck alone explains the E2-8/E2-128-25 gap, since the two share identical encoders: the 8D compression merges the pathways too aggressively for one to compensate when the other is corrupted. Joint corruption confirms the gain is structural.

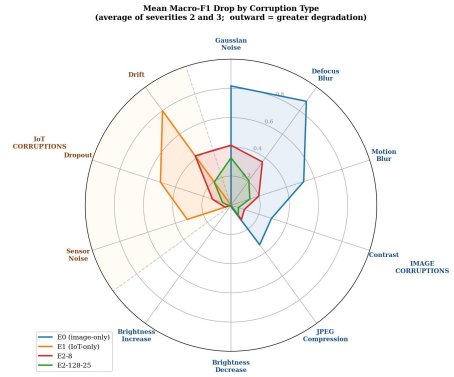


Fig. 5. Mean Macro F1 drop by corruption type (severities 2–3) for E0, E1, E2-8, and E2-128-25; traces nearer the centre are more robust.

C. Discussion

E2-8 achieves higher early-stage recall (98.04%) and accuracy (98.75%) than either unimodal baseline, with the most meaningful gain in Macro F1 (0.9639 $\rightarrow$ 0.9733), the metric most sensitive to minority classes under this imbalance — fusion resolves visually similar diseases whose early stages occur under different humidity/temperature regimes. The 4D collapse shows fusion helps only if the fused representation keeps enough dimensions to preserve stage boundaries (8 sufficed here). The robustness study extends this along two

axes: fusion increases robustness ($E2-8 > E0/E1$), and removing compression increases it further ($E2-128-25 > E2-8$), giving the ordering unimodal $< E2-8 < E2-128-25$. Two takeaways: the fusion result matters because real deployments cannot guarantee clean images or reliable sensors, so graceful degradation is more deployable than dependence on a single clean input; and the width result quantifies the cost of the bottleneck in robustness terms. In sum, the classical fusion experiments produce a strong clean-data classifier and a robustness benchmark on a large, highly imbalanced dataset, with E2-128-25 the most resilient variant and E2-8 the direct comparison baseline.

VI. CONCLUSION AND FUTURE WORK

We implemented and evaluated a multimodal framework for crop disease detection that fuses RGB imagery (MobileNetV3) with IoT sensor telemetry (MLP), outperforming both unimodal baselines in clean-data accuracy and in robustness to test-time corruption—confirming that structural robustness stems from independent modality pathways compensating for one another during channel degradation. Building on these established classical baselines and bottleneck interfaces, our next phase will replace the 8-dimensional bottleneck with a parameterized Variational Quantum Circuit (VQC) to benchmark hybrid quantum-classical computing under real-world agricultural constraints.

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